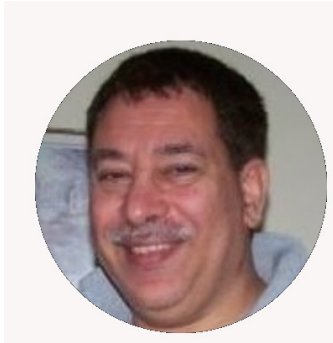


Nicholas (Nico) Michael

| Senior Java Developer Team Lead | Solutions Architect



1. Personal Information

Nationality: South African

Location: Pretoria, Gauteng, South Africa

Languages: English & Afrikaans

Contact Details

Email: nicomichael2018@yahoo.com

Cell: 082 508 1950

YouTube DevOps Channel : <https://www.youtube.com/@dvp7388>

GitHub: <https://github.com/nic0michael>

AI Portfolio video: https://youtu.be/GyjAqyJZw_Y

Professional Summary

Senior Java Developer, Technical Lead and Solutions Architect with **26+ years of Java** development experience and **6 years of architecture experience** across Infrastructure and Solution Architecture.

Delivered 56+ Spring Boot microservices into production at WesBank and worked across Banking, Finance, Healthcare, Insurance, Telecommunications, Education, Loyalty and Utility sectors.

Strong practical experience in **Java, Spring Boot, Kubernetes, Docker, Terraform, Azure AKS** and CI/CD.

Development Best Practice Coach specializing in TDD, BDD, Clean Code, DevOps and AI-assisted development.

Core Competencies

- Software Engineering: Java, **Spring Boot**, **REST** APIs, JPA, Full Stack Development
- Technical Leadership: **Team Leadership**, **Mentoring**, Training, Development Best Practices
- DevOps & Cloud: **Kubernetes**, **Docker**, **Helm**, **Jenkins**, **GitLab CI/CD**, **Terraform**
- Cloud Platforms: **Azure AKS**, **AWS**, **GCP**, **Oracle Cloud**
- Architecture: Solution Architecture, Infrastructure Architecture, Enterprise Architecture
- Cloud-Native: **Kubernetes**, **K3S**, **Docker**, **Helm**, Cloud-Native Architecture
- Software Quality: TDD, BDD, JUnit, Mockito, **SonarQube**, Clean Code
- AI-Assisted Development: Spec Driven Development SDD, Context Driven Development CDD, Prompt Engineering, Multi-Agent Development
- AI Tools: GitHub Copilot, Claude Code, Codex, Cursor
- Process & Architecture: **UML**, **C4**, **TOGAF**, **IBM GS Method**

DevOps & Cloud Engineering Experience

Practical experience across DevOps, cloud infrastructure, automation, containerisation, Infrastructure as Code, CI/CD and Kubernetes.

- 2 years **Azure** experience including Azure AKS Kubernetes and Pipelines.
- 1 year **GCP** experience developing a Notification App using Firebase in GCP.
- 1 year practical **AWS** experience building infrastructure for Udemy training.
- 4 years **Oracle Cloud** provisioning **Rocky Linux** servers
- 8 years **Docker** experience.
- 7 years **Kubernetes** and Cloud-Native platform experience.
- Applied **Terraform** for repeatable Infrastructure as Code deployments.
- Developed PowerShell and Linux shell automation for development, testing, analysis and deployment.
- Worked with **Docker Compose** and **multi-container environments**.
- Practical Kubernetes experience covering K3S, workloads, nodes, namespaces, pods, services and troubleshooting.
- Experience with Proxmox virtualised infrastructure.
- Experience with **SonarQube** for automated code-quality and security analysis.
- Practical troubleshooting of networking, ports, DNS, certificates, containers, virtual machines and deployment failures.
- Applied **DevSecOps** principles within the software delivery lifecycle.

Architecture Experience

Infrastructure Architect: IBM, ESKOM – 3 years

Solution Architect: Cell C, FleetDomain, Vertice – 3 years

Designed and supported enterprise, infrastructure and application architectures across financial services, telecommunications, utilities and healthcare environments.

AI-Assisted Development & Engineering Leadership

Led research, evaluation and adoption of AI-assisted software development tools including GitHub Copilot, Claude Code, OpenAI Codex and Cursor.

- Researched and implemented Spec-Driven Development using Microsoft GitHub Spec Kit.
- Developed corporate AI policy files and coding governance guidelines for AI-generated code.
- Evaluated single-agent and multi-agent development approaches.
- Evaluated competing AI platforms using a common full-stack application and recommended Claude Code and Codex based on practical results.
- Researched local LLM development using Ollama, OpenCode and Google Gemma.
- Created the Iterative Context-Driven Development (ICDD) methodology using AI-assisted brainstorming, requirement refinement and iterative quality reviews.
- Produced technical training content through a DevOps-focused YouTube channel.

Key Achievements

- Delivered **56 Spring Boot microservices** into production at WesBank.
- Designed and implemented a **High Availability K3S Kubernetes cluster** spanning Germany, England and Johannesburg.
- **Developed a Python-based K3S Kubernetes Cluster Generator** to automate Kubernetes cluster configuration and generation.
- Used Helm Charts to automate application deployment and management across Kubernetes.
- Created technical demonstrations **integrating Terraform, Kubernetes, Helm and Proxmox**.
- Introduced TDD, BDD and DevOps best practices across development teams.
- Developed the AHAA Agentic Development Methodology.
- **Lead the integration of the Billing Switch** application into **Azure AKS Kubernetes**.
- Published a Microservices White Paper.
- Published 50 Open Source projects on GitHub, including K3S Auto Cluster Generator and Claude Code Multi Agent With Policies.
- **Provided technical training** in Java, Kubernetes, DevOps and AI-Assisted Development.

Projects

- **WesBank:** Delivered **56 Spring Boot microservices** into production.
- **IBM:** Analyzed and delivered GJMC System Architecture
- **CellC:** Analyzed Synix System made substantial changes to the System Architecture and achieved significant improvement to the System performance
- **Fleetdomain:** Created their Security policy and their automated backup system
- **LoyaltyPlus:** Created their Automated Backup system, Developed their Messaging System, Created a High Availability Kubernetes Cluster
- **Vertice:** Dockerized their Billing Switch project. Led the team that took it to Azure AKS Kubernetes. Implemented their Architectural Decisions Register. Managed their System Architecture Review process

Employment History

Vertice

Solution Architect | Aug 2025 – Present

Focus: Enterprise Solution Architecture, AI Agentic Systems R&D, AI Architecture, Agent-Oriented Design, Technical Strategy and Innovation.

- Maintained the Architecture Intellectual Capital repository in SharePoint.
- Maintained the Architectural Decision Register (ADR) process and repository.
- Maintained the System Architecture Review (SAR) framework and documentation.
- Researched and evaluated emerging AI-assisted software development tools.
- Researched Spec-Driven Development using GitHub Spec Kit.
- Researched Claude Code, Cursor and Codex.
- Investigated single-agent and multi-agent AI development approaches.
- Developed corporate AI policy frameworks and coding governance guidelines.
- Created and maintained the AHAA methodology, drawing on C4, TOGAF, Zachman and IBM Global Services methodologies.
- Developed Tech-Radar applications providing real-time technology adoption information for management.
- Developed applications providing management with Code Quality, Security Issues and Technical Debt information, project managers with Git repository status, and business analysts with development task status.
- Provided training on Agentic Development tools.

Tech Stack: GitLab, Azure DevOps Repo, Azure AKS Kubernetes, GitHub Copilot, GitHub Spec Kit, Claude Code, Cursor, Codex, VS Code, IntelliJ, Java, Python, Angular, Spring Boot

Vertice

Senior Java Developer | Full Stack Developer | Best Practice Coach | Aug 2024 – Aug 2025

- Developed application code and provided technical training to development teams.
- Led the Dockerization of the Billing Switch.
- Mentored team members in Docker adoption.
- Participated in the Azure AKS Kubernetes POC.
- Guided the team through its first AKS Kubernetes adoption and integration.
- Provided Kubernetes and cloud-native development training.
- Introduced and delivered Helm training to the DevOps team.
- Trained development teams in TDD.
- Introduced and trained teams on Spec-Driven Development.
- Taught Context-Driven Development using Claude Code.
- Introduced advanced brainstorming techniques to improve SDD and CDD outcomes.

Tech Stack: Java, Spring Boot, Docker, Dockge, Kubernetes, K3S, Azure AKS, Helm, GitLab, Azure DevOps

LoyaltyPlus

Senior DevOps Consultant / Senior Developer | Jun 2023 – Jul 2024

- Designed and **implemented Disaster Recovery** and backup strategy for customer WordPress platforms using Ansible, Docker, HAProxy and Semaphore.
- Built and automated **High Availability K3S Kubernetes cluster** across Germany, England and South Africa.
- Developed **infrastructure automation using Ansible, Terraform, Proxmox**, Cockpit and Rocky Linux.
- Developed a distributed messaging platform using Camel Quarkus, Kafka, Angular and REST APIs supporting SMS, Email, WhatsApp, Push Notifications and In-App Notifications.
- Implemented Firebase integration for mobile notification services.
- Designed security monitoring and vulnerability scanning using Metasploit and StarRocks.
- Evaluated and implemented Zero Trust networking using NetBird.

Tech Stack: Java, Spring Boot, **Camel, Quarkus, Kafka, Angular, Docker, K3S Kubernetes, Helm, Ansible, Terraform, Proxmox, HAProxy, Semaphore, NetBird, Metasploit, StarRocks, Google GCP, Firebase**, REST APIs, Rocky Linux Servers.

Advance / MTN

Senior Software Engineer | Kubernetes Specialist | Technical Trainer | Jun 2022 – Jun 2023

- Developed solutions for the MTN USSD platform.
- Designed and implemented **High Availability K3S Kubernetes clusters** on Red Hat infrastructure.
- Created Kubernetes manifests and Helm Charts for Zookeeper, Elasticsearch, Kibana, Prometheus, PostgreSQL and MongoDB.
- Led Kubernetes adoption and deployment practices across development teams.
- Delivered training in TDD, JUnit, Mockito, Kubernetes, Helm and cloud-native deployment practices.

Tech Stack: Java, Spring Boot, **Kubernetes**, K3S, **Helm**, **Docker**, Elasticsearch, Kibana, **Prometheus**, PostgreSQL, MongoDB, Zookeeper, JUnit, Mockito, **Jenkins**, Terraform

DVT

Senior Java Developer Team Lead | Feb 2017 – Jun 2022

Discovery | 2022

- Led Spring Boot Micro-service development initiatives.
- Developed Java training curricula and technical enablement programs.

Tech Stack: Java, Spring Boot Micro-services, Hibernate, GitLab, TDD, Mockito.

Curro Schools | 2021

- Served as Software Development Best Practices Coach and Senior Java Developer.
- Delivered Proofs of Concept covering TDD, BDD, **Docker**, **SonarQube**, Portainer, RabbitMQ, Postman and Micro-service architecture.
- Trained development and QA teams in automated testing and engineering best practices.

Tech Stack: Java, Groovy, Grails, Spring Boot Micro-services, **OAuth**, TDD, **SonarQube**

WesBank | 2018 – 2020

- Delivered 56+ Spring Boot microservices into production.
- Led adoption of TDD, BDD, **SonarQube**, **Bitbucket** and Confluence.
- Designed a KYC Rules Engine using Drools and RabbitMQ.
- Architected integration services for LDAP, App Notifications, CT02, CIS, Actimize and Documentum.
- Produced solution designs, architectural artifacts and technical documentation.
- Received **Kubernetes training** from WesBank
- Completed **AWS** training through Udemy and online practical for Just over 1 Year .

Tech Stack: Java, **Spring Boot**, **SOAP**, **RabbitMQ**, **Docker**, Docker Compose, Portainer, **SonarQube**, **Postman**, JUnit, Mockito, TDD, BDD, **Drools**, **Kubernetes**, Elasticsearch, Logstash, Kibana, LDAP, IBM ODM, **Nexis**.

Sanlam | 2020

- Developed Spring Boot microservices integrated with IBM ODM Business Rules Management Engine.
- Used IBM Rule Designer to develop process flow diagrams.

Tech Stack: Java, **Spring Boot**, JUnit, TDD, **IBM ODM**, GitLab

Momentum | 2017 – 2018

- Was the Full-stack Developer Team Lead
- Responsible for architecture, TDD adoption, code quality.
- Worked with **Docker**, **Docker Swarm** and **Ansible** automation

Tech Stack: Java, **Angular 2**, **Typescript**, **Bootstrap**, **Docker**, Docker Swarm, Postman, JUnit, Mockito, TDD, IBM ODM, Ansible, GitLab

FleetDomain

Solutions Architect | Infrastructure Architect | Senior Java Developer | May 2016 – Jan 2017

- Implemented enterprise security policies, disaster recovery, backup automation and UAT environments.
- Designed High Availability production architecture and hot backup solutions.
- Reverse-engineered and optimised Groovy/Grails enterprise applications.
- Modernised deployment architecture and server provisioning processes.

Tech Stack: Java, Groovy, Grails, Ubuntu, Windows Server, Linux Administration, Disaster Recovery, Security Architecture, Backup Automation

Standard Bank

Senior Java Developer | Jul 2015 – May 2016

- Developed enterprise OTP authentication services for Africa and Offshore Banking platforms.
- Delivered JEE8 interceptor framework and Vasco OTP integration.
- Applied TDD and BDD using JUnit, Mockito, WireMock, JBehave and Serenity.

Tech Stack: Java, JEE8, **SOAP**, **Maven**, JUnit, Mockito, WireMock, JBehave, Serenity BDD, SOAP, Web Services. Centos Linux, Puppet.

Cell C

Java Solution Architect | Infrastructure Architect | Senior Java Developer | Mar 2014 – Jul 2015

- Redesigned enterprise systems for scalability and performance.
- Changed the Architecture of Badly performing system
- Analyzed the Siemens Nokia Synix architecture Produced UML models.
- Optimized the Complete System Design.
- Made substantial improvement to performance

Tech Stack: Java, UML, Enterprise Architect, EJB, Oracle SQL, Centos Linux, Percona MySQL, JUnit, TDD, Javascript.

Signio

Senior Java Developer | Nov 2009 – Feb 2014

- Developed Standard Bank integration gateways and insurance platform integrations.
- Delivered Jasper Reports solutions and Groovy/Grails applications.
- Developed a project for the VAF5 System to send requests to the NATIS gateway
- Delivered new gateways to the Insurance companies

Tech Stack: Java, EJB3, SOAP, Centos Linux, Percona MySQL, Groovy, Grails, Jasper Reports

Paracon

ESKOM

Infrastructure Architect | Senior Java Developer | Apr 2007 – Jul 2009

- Delivered architecture and integration designs for the Online Vending System
- Installed and configured enterprise integration platforms and middleware.
- Developed BPEL Business Processes for SOAP integration connecting SAP using Sun Java CAPS.
- Worked as **Infrastructure Architect** for 2 years.

Tech Stack: Java, **SOAP, BPEL, LDAP**, Sun Java CAPS, Infrastructure Architecture, **Ubuntu** Linux Server, **VmWare ESX Server**

ABSA Bank

Senior Java Developer | Jan 2006 – Mar 2007

- Developed Home Loans portal solutions using JSR168 Portlets and JSF.
- Participated in delivery of the production Home Loans platform.

Tech Stack: Java, JSR168 Portlets, JSF

IBM Global Services

Infrastructure Architect | Java Developer | Feb 1998 – Jun 2002

- Developed components of the National Online Casino Gambling Monitoring System.
- Worked as Infrastructure Architect.
- Delivered **Infrastructure Architecture for StanLib project**
- Created Architectural artifacts for the GJMC project for Johannesburg Metro Council
- I conducted the **CMM Maturity Rating of Rubico** for the World Bank
- Was part of the IBM **GS Method Methodology** training team of IBM

Tech Stack: Java, Java EJB, Java J2EE, RUP, Visual Age for Java, UML Sybase SQL server, IBM DB2, Windows NT Server, O/S2

Education, Professional Training & Qualifications

National Higher Diploma – Production Management

Tshwane University of Technology | Graduation 1989

National Diploma – Electrical Engineering L/C

University of Johannesburg | Graduation 1980

Matric

TED | 1975

Professional Training:

SQL,Java, EJB JEE, UML, C++, Kubernetes, Docker, Jenkins, Terraform, Azure, AWS, Metasploit and Agile Methodologies.

Training providers include **IBM, Damelin and Udemy**

Publications, Open Source & Community Contributions

Published Microservices White Paper on Test Driven Development.

- <https://www.dvt.co.za/white-paper-building-cloud-ready-microservices-using-industry-best-practices>

Published 50 Open Source projects on GitHub.

- Created the K3S Kubernetes Auto Cluster Generator
- The Claude Code Multi Agent Development project With Policies.
- Technical training in Java, Kubernetes, DevOps and AI-Assisted Development.

Published Videos in DevOps YouTube channel

- Videos covering Java, Enterprise Java, **Quarkus**, TDD, **Docker**, **Kubernetes**, Portainer, **Proxmox**, **Terraform**/OpenTofu, **Helm**, **Jenkins**, **Ansible**, AI Agentic Development and N8N.

Professional Memberships

- SAIETE – South African Institute for Technician Engineers
- SAIEE – South African Institute for Electrical Engineers
- JUGSA – Java User Group of South Africa
- GLUG – Gauteng Linux User Group

References

Professional and personal references available upon request.